Student Guide — middle / module-07-eleven-multiplication

IKS Curriculum

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Student Workbook

Student Workbook — Module 7 Eleven Multiplication (Middle)

Name: ____________________ Class: ____ Section: ____

This is your workbook for the next two weeks. Bring it to every class.

Day 1 — Baseline

My baseline time for 73 × 11 (long multiplication): ____ seconds

My first try with the ×11 trick: ____ seconds (mostly worked? Y / N)

The trick (in my own words): 1. ____ 2. ____ 3. ____

Day 2 — The Carry Case

When the digit-sum is ≥ **10**: - Middle digit = ______________________ - Carry the ___ into the ______________________

Worked example (mine): _____ × 11 = _____ (digit-sum = ___)

Day 3 — Why It Works

The algebra (in my handwriting):

```
(10a + b) × 11
    =                            (distribute)
    =                            (rearrange)
    =  100a + 10(a+b) + b        (final form)
```

In one sentence: the trick works because ________________________

Day 4 — Three-Digit × 11

The pattern for *abc* × 11: ______________________________________

Worked example: 234 × 11

Thousands: ___ Hundreds: ___ Tens: ___ Units: ___ → ______

Day 5 — Mental-Math Speed Day

Day 1 baseline (73 × 11, long way): ____ sec Day 5 retest (73 × 11, with trick): ____ sec **% improvement:** _____

My project choice: ________________ (video / workbook / proof poster) Partner: ________________________

Day 6 — *Anurūpyeṇa*

For 2-digit *ab* × (10 + *k*): - units = ________________________ - tens = ________________________ - hundreds = ________________________

Example: 45 × 13 = _______

Day 7 — When the Trick Doesn't Help

The ×11 trick (and its scaled cousins) works for multipliers of the form _____________.

For multipliers far from 10, use ________________________.

Day 8 — Tirthaji's Sūtra System

Vedic Mathematics (Tirthaji) was published in the year _____ (___ years after Tirthaji died).

Tirthaji claimed the sūtras come from an __________________ (a Vedic appendix text).

The mathematician ______________ (Dani / Plofker / ?) wrote a critique in _____.

My current view: ____

Day 9 — Project Progress

What I worked on today: ____

What's still left: ____

Day 10 — Reflection

1. What was the most surprising thing you learned about ×11?

2. Did your view of "Vedic mathematics" change during this module? How?

3. If you had three more days, what would you do differently?

Glossary (your own)

Sanskrit	Meaning	First met
Ekādhikena Pūrveṇa	"by one more than the previous"	Day 1
Anurūpyeṇa	"proportionally"	Day 6
Tirthaji	name (1884–1960)	Day 8
sūtra	aphorism / formula	Day 8
Pariśiṣṭa	appendix / supplement	Day 8

My favourite ×11 problem from the module

(Write the problem and your worked answer. Use the dot trick to show off.)

Problem: _____ × 11 = _____

Why I like it: ____

Day 1 — The Hook: *2 _ 3 = 253*

Module: ×11 (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will compute any 2-digit × 11 product where the digit-sum is less than 10, in under 3 seconds, and will have recorded a personal baseline time for the same problem using standard long multiplication.

Materials

- Printed baseline sheet: ONE multiplication problem (73 × 11). One per student.
- Stopwatch or wall clock with a second hand. (One per pair is plenty.)
- Whiteboard / large paper to demonstrate.

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: ekādhikena pūrveṇa; lit. "by one more than the previous") — the Sanskrit name Tirthaji (1965) gives the underlying sūtra. We use it as the cultural anchor; the technique is what students take home.

Lesson flow

Warm-up (5 min)

- Teacher: *"How long does it take you to compute 73 × 11 the normal way? Make a guess."*
- Take 3–4 guesses. Write the range on the board.

Baseline timing (8 min)

- Hand out the baseline sheet face-down. Students write their name and predicted time on the back.
- On "Go," flip and solve 73 × 11 using long multiplication.
- Record actual time honestly. Mark answer themselves.
- Expected range: 15–60 seconds.

Teacher demonstration (12 min) — The trick

- Write on the board:

  ```
  73 × 11 = ?
  ```

- *"I'm going to show you a method that takes about 2 seconds, then we'll figure out why it works."*

```
Step 1: Write the digits, with a gap:    7  _  3
Step 2: Add the two digits: 7 + 3 = 10.
                              Wait — that's the carry case. Park it.
```

- *Reset. Different number first.* Try **23 × 11**:

```
Step 1: Split:    2  _  3
Step 2: Add:      2 + 3 = 5
Step 3: Drop in:  2  5  3   →   253.
```

  Check on the board with long multiplication: 23 × 11 = 23 × 10 + 23 = 230 + 23 = 253. ✓

- Try **45 × 11**:

```
4 _ 5  →  4 + 5 = 9  →  4 9 5  →  495.
```

- Try **62 × 11**:

```
6 _ 2  →  6 + 2 = 8  →  6 8 2  →  682.
```

- *"Pattern: outer digits stay the same; middle digit = their sum."*

- Now name it. *"Tirthaji (1965) calls this **Ekādhikena Pūrveṇa** — 'by one more than the previous.' That is the Sanskrit name of the sūtra. The trick itself is just (10a + b) × 11 = 100a + 10(a+b) + b, which we'll prove on Day 3."*

Activity (15 min) — Try it yourself

- Worksheet (handed out): 10 two-digit × 11 problems, ALL with digit-sum < 10.

```
12 × 11    23 × 11    34 × 11    45 × 11    52 × 11
61 × 11    16 × 11    27 × 11    35 × 11    44 × 11
```

- Pair work. Partner times each other. Goal: each problem under 3 seconds.

- After 10 min, students re-do **73 × 11** on the back of the baseline sheet. *Hmm — 7 + 3 = 10. That's tomorrow's lesson.*

Wrap-up (5 min)

- Show of hands: who finished all 10 in under 30 seconds? (Most will.)
- Tease tomorrow: *"73 × 11 broke the rule because 7 + 3 = 10 and you can't write 10 in the middle of a number. Tomorrow — the carry case."*

Student-facing content

The ×11 Trick (no-carry case)

For any 2-digit number with digit-sum < 10: 1. **Split** the digits with a gap. 2. **Add** the two digits. 3. **Drop** the sum into the gap.

Example: 34 × 11 → 3 _ 4 → 3+4 = 7 → 374.

Tirthaji (1965) names the underlying sūtra ***Ekādhikena Pūrveṇa*** — “by one more than the previous.” The trick is older than its modern Sanskrit name; the algebra (which we’ll do on Day 3) is the same in every language.

Homework

- Find any two 2-digit numbers (digit-sum < 10) in your house — phone numbers, page numbers, prices. Compute each × 11 using the trick. Bring 5 worked examples to class tomorrow.

Differentiation

- **Tier up:** Try 3-digit × 11 (e.g. 234 × 11, 152 × 11). See if you can guess the pattern before Day 4.
- **Tier down:** If the gap is confusing, write the answer as a 3-digit number directly: hundreds = first digit, tens = sum, units = second digit.

Teacher notes

- **The baseline time matters.** Day 5’s retest depends on having a Day 1 record.
- Don’t dwell on the Sanskrit name today. *Ekādhikena Pūrveṇa* is the cultural anchor; the win is mental speed.
- Some students will already know the trick (especially those from competitive-math backgrounds). Have them be peer-coaches. Reframe their advantage as “you’ll learn the WHY this week.”
- About 30% of randomly-chosen 2-digit numbers have digit-sum ≥ 10 (the carry case). So today’s worksheet hand-picks the no-carry ones; tomorrow we add the carries.

Citation(s) used in this lesson

- Tirthaji, B. K. (1965). *Vedic Mathematics*. Motilal Banarsidass. Sūtra 1, *Ekādhikena Pūrveṇa*.

Day 2 — The Carry Case

Module: ×11 (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will correctly handle the carry case in 2-digit × 11 (where the digit sum ≥ 10), including problems like 87 × 11 = 957, in under 5 seconds.

Materials

- Whiteboard.

- Printed worksheet of 12 problems, mixing no-carry and carry cases.
- Stopwatch.

Lesson flow

Warm-up (5 min)

- Neighbour-sum drill: teacher calls a 2-digit number with digit-sum < 10; class chorus the ×11 product.
- *“12 times 11!”* → *“132!”* — *“34 times 11!”* → *“374!”* — 10 in a minute.

Recall (3 min)

- Yesterday’s problem: 73 × 11. 7 + 3 = 10. Why was this awkward?
- *“You can’t put 10 in the middle of a 3-digit number. So what do we do? Carry the 1 — exactly like in standard addition.”*

Core (15 min) — The carry case

1. **Demonstrate 73 × 11 fully:**

   ```
   Step 1: Split:    7  _  3
   Step 2: Add:      7 + 3 = 10
   Step 3: Write the units digit of the sum (0) in the middle:    7
   0  3
   Step 4: Carry the tens digit of the sum (1) into the leftmost
   digit: 7 + 1 = 8.
   Step 5: Answer:   8  0  3   →   803.
   ```

   Check: 73 × 11 = 73 × 10 + 73 = 730 + 73 = **803**. ✓

2. **Try 87 × 11:**

   ```
   8  _  7
   8 + 7 = 15
   Middle = 5; carry 1 into the 8 → 9.
   Answer: 9  5  7 → 957.
   ```

   Check: 87 × 11 = 870 + 87 = **957**. ✓

3. **Try 99 × 11:**

   ```
   9 _ 9
   9 + 9 = 18
   Middle = 8; carry 1 into the 9 → 10.
   But 10 isn't a single digit — so it becomes a "1" carried
   further: leftmost = 0, with a 1 carried to the left of that.
   Answer: 1  0  8  9 → 1089.
   ```

   Check: 99 × 11 = 990 + 99 = **1089**. ✓ (4-digit answer because the carry cascaded.)

4. **The full procedure** (boxed on the board):

For any 2-digit number *ab* × 11:

1. Compute $a + b$.
2. If the sum is < 10: answer is $a\ (a+b)\ b$.
3. If the sum is ≥ 10: middle digit = units digit of sum; add 1 to a.
4. If the new a is ≥ 10 (only happens at 99): cascade the carry.

Activity (17 min) — Mixed practice

- Worksheet of 12 problems, half no-carry, half carry:

```
21 × 11    35 × 11    47 × 11    58 × 11
62 × 11    74 × 11    83 × 11    89 × 11
91 × 11    99 × 11    18 × 11    27 × 11
```

- Round 1 (8 min): solo, untimed. Self-check with long multiplication on any 3 of them.

- Round 2 (8 min): paired. Partner calls a problem; you answer aloud. Goal: under 5 seconds per problem.

Wrap-up (5 min)

- Quick chorus: *"What does the trick break into?"* — Split, add, drop (and carry if needed).
- Tease Day 3: *"Tomorrow we PROVE why this works. You'll write the algebra yourself."*

Student-facing content

The ×11 Trick (with carry)

1. Split the digits.
2. Add them.
3. **If the sum < 10:** drop it in the middle. Done.
4. **If the sum ≥ 10:** keep the units digit of the sum as the middle; add 1 to the leftmost digit.

Examples: - 45 × 11: 4+5=9 → 4**9**5 = **495** - 87 × 11: 8+7=15 → 8**(1)5**7 → **9**57 = **957** - 99 × 11: 9+9=18 → 9**(1)8**9 → cascade → 1**0**89 = **1089**

Homework

- 8 mental ×11 problems (carry case): 76, 84, 85, 88, 93, 96, 67, 78 — multiplied by 11. Write the answers.
- Time yourself. Goal: under 30 seconds total.

Differentiation

- **Tier up:** Find a 2-digit number where the trick gives a 4-digit answer (i.e. 99 × 11 = 1089). Can you find any others, or is 99 unique? Justify.
- **Tier down:** Solve only the no-carry problems on the worksheet today. We'll cement carries on Day 4.

Teacher notes

- The most common error: students remember to write the units digit of the middle sum but **forget** to add the carry into the leftmost digit. Watch for "73 × 11 = 7**0**3" with no carry added — this is a classic mistake.
- The 99 × 11 case (and the cascading carry) genuinely surprises students. Lean into it — it's evidence the algebra is doing real work, not arithmetic by coincidence.
- If a student asks "why does this work?" — that's Day 3. Park it. Honour the curiosity by writing the question on a board sticky.

Citation(s) used in this lesson

- Tirthaji (1965), sūtra 1 (*Ekādhikena Pūrveṇa*) — the technique name.
- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center) — worked examples of the carry case.

Day 3 — Why It Works

Module: ×11 (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will derive the identity $(10a + b) \times 11 = 100a + 10(a + b) + b$ and explain in their own words why the ×11 trick is not a coincidence.

Materials

- Whiteboard.
- Worksheet: "explain to a friend" prompt with space for student algebra.

Lesson flow

Warm-up (5 min)

- Neighbour-sum chorus: 15 quick ×11 problems, mix of carry and no-carry.
- *"By now your hands know the trick. Today your brain catches up."*

Core (20 min) — The algebra

1. **Write any 2-digit number as 10*a* + *b*.** Make sure every student gets this.

   ```
   23 = 10·2 + 3   →  a = 2, b = 3
   87 = 10·8 + 7   →  a = 8, b = 7
   ```

2. **Distribute the multiplication by 11.** Step by step on the board, students copy:

   ```
   (10a + b) × 11
       = (10a + b) × (10 + 1)
       = (10a + b) × 10  +  (10a + b) × 1
       = 100a + 10b      +  10a + b
   ```

```
= 100a + 10a + 10b + b
= 100a + 10(a + b) + b
```

3. **Read the final line out loud.** *“100a + 10(a+b) + b. What does this say in plain language?”*

    - $100a$ — the **hundreds** digit is a (the original first digit).
    - $10(a+b)$ — the **tens** digit is $a + b$ (the digit sum).
    -
        - b — the **units** digit is b (the original second digit).

    “That IS the trick. The algebra spells out: first digit, sum, last digit.”

4. **Check on 23:**

    ```
    a = 2, b = 3.
    100·2 + 10·5 + 3 = 200 + 50 + 3 = 253. ✓
    ```

5. **Now the carry case** — where does the carry come from?

    ```
    a = 8, b = 7.
    100·8 + 10·15 + 7 = 800 + 150 + 7 = 957.
    ```

    “Notice: 10 × 15 = 150 = 100 + 50. The 100 contribution slides into the hundreds column — that IS the carry. So the algebra ALSO explains why we add 1 to the leftmost digit when the sum is ≥ 10.”

Activity (15 min) — Explain to a friend

- Worksheet prompt:

    Imagine a Grade 4 cousin asks: *“Why does the ×11 trick work? Isn’t it just magic?”* Write 5–7 sentences explaining why it’s NOT magic. Use the words **distribute**, **digit sum**, and **place value**.

- 10 min individual writing.
- 5 min pair-sharing: read your paragraph to your partner; they tell you one part that’s clear and one part that’s unclear.

Wrap-up (5 min)

- Read aloud one strong paragraph (with permission).
- Preview Day 4: *“Tomorrow — extend the trick to 3-digit numbers. The algebra will scale.”*

Student-facing content

Why the ×11 trick works

Write any 2-digit number as $10a + b$ — the first digit is worth 10, the second is worth 1.

Distribute the multiplication by $11 = 10 + 1$:

$(10a + b) \times 11 = (10a + b) \times 10 + (10a + b) \times 1 = 100a + 10b + 10a + b = \mathbf{100a + 10(a + b) + b}$.

In plain language: **hundreds = *a*, tens = (*a* + *b*), units = *b*.** That's the trick.

When $a + b \geq 10$, the "$10(a+b)$" term overflows the tens place by exactly one hundred — that's the carry.

It is not magic. It is the distributive law.

Homework

- Apply the proof to a number you choose. Pick any 2-digit number, write it as $10a + b$, distribute by 11, and show the final form. Show all 5 lines.

Differentiation

- **Tier up:** Prove the same identity for ×111 (multiplication by 111). What's the equivalent "split-and-add" pattern? Test it on 23 × 111.
- **Tier down:** Don't worry about the carry-case derivation. Get the no-carry version of the proof working.

Teacher notes

- This is the conceptual core of the module. Don't rush.
- Students who can write the algebra but can't *speak* it should be pushed to verbalise — "distribute" is the verb that matters here.
- The most common error in student paragraphs: confusing "the trick" with "the answer." The trick is the algorithm; the algebra justifies the algorithm.
- About 20% of students will object: *"But this is just multiplication, not anything special."* That's the right reaction. Honour it: *"You're right — and that's the deeper point. The 'special' name (Ekādhikena Pūrveṇa) is a label; the math underneath is regular algebra."*

Citation(s) used in this lesson

- (None required for the algebra itself — it is original derivation.)
- Optional teacher reference: Tirthaji (1965), sūtra 1, for the framing.

Day 4 — Three-Digit × 11

Module: ×11 (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will compute any 3-digit × 11 product using the running "neighbour-sum" pattern, including carry cases, and will complete Part A of the formative quiz.

Materials

- Whiteboard.

- Printed worksheet of 10 three-digit × 11 problems.
- Formative quiz Part A (`quizzes/formative.md`).

Lesson flow

Warm-up (5 min)

- Neighbour-sum chorus: 10 quick 2-digit × 11 problems including carry cases. (Should be near-instant by now.)

Core (18 min) — The 3-digit pattern

1. **Set up.** *"Yesterday we proved (10a+b)×11. What happens with a 3-digit number 100a + 10b + c?"*

2. **Derive together** (or have students derive):

   ```
   (100a + 10b + c) × 11
       = (100a + 10b + c)(10 + 1)
       = 1000a + 100b + 10c + 100a + 10b + c
       = 1000a + 100(a + b) + 10(b + c) + c
   ```

3. **Read the result.** *"Thousands = a. Hundreds = a + b. Tens = b + c. Units = c."*

   In words: **first digit stays, then each pair of adjacent digits sums and slides into the gap, then the last digit stays.**

4. **Worked example: 234 × 11.**

   ```
   Splits and gaps:    2  _  _  4
                          ↑     ↑
                        (a+b) (b+c)
   2 + 3 = 5
   3 + 4 = 7
   Answer:             2  5  7  4   →   2574.
   ```

   Check: 234 × 11 = 234 × 10 + 234 = 2340 + 234 = **2574**. ✓

5. **Worked example with carry: 567 × 11.**

   ```
   5  _  _  7
   5 + 6 = 11   ← carry case
   6 + 7 = 13   ← carry case
   ```

   Strategy: work **right to left** to handle carries properly.

   ```
   Units: 7.
   Tens: b + c = 6 + 7 = 13. Write 3, carry 1.
   Hundreds: a + b + carry = 5 + 6 + 1 = 12. Write 2, carry 1.
   Thousands: a + carry = 5 + 1 = 6.
   Answer:    6  2  3  7   →   6237.
   ```

   Check: 567 × 11 = 5670 + 567 = **6237**. ✓

6. **Generalisation.** For an *n*-digit number, the trick is the same pattern:
    - Outer digits stay.
    - Every interior digit becomes the sum of the two original digits flanking that position.
    - Carries cascade from right to left.

Activity (15 min) — Worksheet

- 10 problems mixing 3-digit no-carry and carry cases:

```
123 × 11    234 × 11    345 × 11    456 × 11    142 × 11
365 × 11    478 × 11    529 × 11    687 × 11    818 × 11
```

- Pair work, self-check by long multiplication on any 3.

Formative Quiz Part A (5 min)

- Distribute `quizzes/formative.md` Part A.
- 5 quick problems, mixed 2- and 3-digit × 11. Closed-notebook.

Wrap-up (2 min)

- Preview Day 5: *"Tomorrow — speed day. We'll retest your Day 1 problem (73 × 11) using the trick."*

Student-facing content

3-digit × 11

For a number *abc* × 11: 1. Write *a* _ _ *c* (outer digits unchanged). 2. Fill the first gap with $a + b$. 3. Fill the second gap with $b + c$. 4. Handle carries right to left.

Example: 234 × 11 → 2 _ _ 4 → 2 (2+3) (3+4) 4 → 2574.

The algebraic identity: $(100a + 10b + c) \times 11 = 1000a + 100(a+b) + 10(b+c) + c$.

Homework

- Solve 10 three-digit × 11 problems on the back of today's worksheet. Time yourself — goal: under 90 seconds total.

Differentiation

- **Tier up:** Try a 4-digit × 11 problem. Derive the pattern first (the identity will give you 5 terms). Test on 2345 × 11.
- **Tier down:** Stay on 2-digit × 11 problems with carries; build speed there before adding the 3-digit layer.

Teacher notes

- The right-to-left direction for carry handling is the key procedural insight. Some students will try left-to-right and run into trouble; redirect them.

- For 567 × 11, the cascading carries (13 in tens, 12 in hundreds) are dramatic. Use that example.
- The pattern is sometimes taught with "underline carries" notation; we use right-to-left because it generalises better to *n*-digit numbers.

Citation(s) used in this lesson

- *Vedic_Maths.pdf* (ISKCON Desire Tree) — practice problems.
- Tirthaji (1965), sūtra 1.

Day 5 — Mental-Math Speed Day

Module: ×11 (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will retest their Day 1 baseline problem (73 × 11) using the trick, complete a paired speed race, and finish Part B of the formative quiz.

Materials

- Day 1 baseline sheet (returned to each student).
- Stopwatch.
- `activities/activity-01-speed-race.md` worksheet.
- Formative quiz Part B (`quizzes/formative.md`).

Lesson flow

Warm-up (3 min)

- Quick chorus: 12 problems, mixed 2- and 3-digit × 11. Cover the carry case.

Day 1 retest (5 min)

- Return Day 1 baseline sheet to each student.
- Re-solve **73 × 11** using the trick. Time yourself.
- Record:

```
Day 1 time (long multiplication): ____ sec
Day 5 time (×11 trick):           ____ sec
Improvement:                      ____ %
```

- Most students will see >70% reduction (e.g. 25 sec → 5 sec).

Activity (20 min) — Paired speed race

See `activities/activity-01-speed-race.md` for the full set-up.

- Round 1 (5 min): Solo, 20 mixed-difficulty problems. Each problem on a flashcard.

- Round 2 (5 min): Paired. Partner reads problem; you answer aloud.
- Round 3 (5 min): Group chorus — teacher displays 10 problems; class shouts the answer.
- Round 4 (5 min): Quick-fire — teacher calls a 2-digit number; first student to call back the ×11 product gets a tally mark. (Light competition; no cumulative score across days.)

Formative Quiz Part B (15 min)

- Distribute Part B.
- 8 questions, mixed 2- and 3-digit × 11, plus 2 short-answer "explain why" items.

Wrap-up (2 min)

- Preview Day 6: *"Tomorrow — the trick extends to ×12, ×13, … all the way to ×19. The middle-add step scales by a multiplier."*

Student-facing content

> **Speed Day**
>
> Today's question: how much faster has your mental ×11 gotten in 4 days?
>
> Record your Day 5 retest time. Compare with Day 1. The gap is the proof of practice.
>
> By end of class you should hit roughly: - 2-digit × 11 (no-carry): under 2 seconds - 2-digit × 11 (carry): under 4 seconds - 3-digit × 11: under 8 seconds

Homework

- 25-problem mixed drill sheet. Solo, untimed. Target: 100% correct.

Differentiation

- **Tier up:** Try 4-digit × 11 problems. Sample: 1234 × 11, 2345 × 11, 4567 × 11. Use the right-to-left carry-cascade method.
- **Tier down:** Focus only on the 2-digit drill today. Build fluency before adding 3-digit.

Teacher notes

- The Day 1 → Day 5 contrast is the emotional payoff of the module. Make sure every student records it; the personal evidence drives motivation.
- Quick-fire (Round 4) should NOT escalate into pure competition. Use it lightly; rotate "first responder" so no single student dominates.
- Some students will plateau on the carry case. Pair them with a peer-coach in Round 2 specifically.

Citation(s) used in this lesson

- (No new sources today — applied practice of Days 1–4.)

Day 6 — *Anurūpyeṇa*: Extending to ×12, ×13, …, ×19

Module: ×11 (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will apply Tirthaji's *Anurūpyeṇa* sub-sūtra ("proportionally") to multiply any 2-digit number by 12, 13, …, 19, with at least 80% accuracy on a mixed worksheet.

Materials

- Whiteboard.
- `activities/activity-02-anurupyena.md` worksheet.
- Stopwatch.

Vocabulary introduced today

- *Anurūpyeṇa* (IAST: anurūpyeṇa; lit. "proportionally") — a sub-sūtra in Tirthaji's corpus. We use it as the name for the proportional scaling that extends the ×11 pattern to ×12, ×13, …, ×19.

Lesson flow

Warm-up (5 min)

- Recall: state the ×11 trick in one sentence.
- *"Outer digits unchanged; middle is the digit sum."*

Core (18 min) — The scaling pattern

1. **Question.** *"What does ×12 look like? Try 23 × 12 by long multiplication first."*

   ```
   23 × 12 = 23 × 10 + 23 × 2 = 230 + 46 = 276.
   ```

2. **Derive.** *"Now distribute (10a + b) × 12 = (10a + b)(10 + 2):*

   ```
   = 100a + 10b + 20a + 2b
   = 100a + 10(b + 2a) + 2b
   ```

   Hmm — that's awkward. *2b* in the units, *(b + 2a)* in the tens. Let's organise it more cleanly.

3. **The Tirthaji pattern.** For N in {11, 12, …, 19}, write $N = 10 + k$ where $k \in \{1, 2, \ldots, 9\}$. Then:

   ```
   (10a + b) × (10 + k)
       = 100a + 10b·k + 10ka + bk
                Wait — let's redo carefully:
       = (10a + b) × 10 + (10a + b) × k
       = 100a + 10b + 10ak + bk
       = 100a + 10(b + ak) + bk
   ```

   Hmm — *bk* could be 2 digits. The pattern is: **scale each digit by k as you slide along.**

   Easier form for students: For $\times N = \times(10 + k)$:

- Start with the leftmost digit.
- Multiply it by *k*, add the next digit, write the units of that sum (carry as needed).
- Continue right.
- The rightmost step: multiply the last digit by *k*.

This is the **Tirthaji form of the *Anurūpyeṇa* extension**.

4. **Worked example: 23 × 13. (k = 3.)**

```
Set up: 0  2  3  (pad with 0 on left to track the carry-out)
Right to left:
  - Units: last digit × k = 3 × 3 = 9. Write 9.
  - Tens: middle digit × k + next-right digit = 2 × 3 + 3 = 9.
Write 9.
  - Hundreds: leftmost digit × k + next-right = 0 × 3 + 2 = 2.
Wait — we want the LEFTMOST as just itself with the carry...
```

(Pause — this is getting confusing. Let's use a different presentation.)

5. **Cleaner presentation for ×12 to ×19.**

For N × 12 of a 2-digit number *ab*: simply use long multiplication mentally, but in the order:

- units of answer = $b \times 2$ (carry if ≥ 10)
- tens of answer = $a \times 2 + b$ (plus carry; carry if ≥ 10)
- hundreds of answer = a (plus carry)

For N × 13: replace ×2 with ×3 everywhere.

General: **For ×(10 + k), multiply each digit by k and add the digit to its right (the "running scan").**

6. **Examples** (boxed):

```
23 × 12:
  units: 3·2 = 6
  tens: 2·2 + 3 = 7
  hundreds: 2 (no carry)
  → 276 ✓

23 × 13:
  units: 3·3 = 9
  tens: 2·3 + 3 = 9
  hundreds: 2
  → 299 ✓

45 × 13:
  units: 5·3 = 15 → write 5, carry 1
  tens: 4·3 + 5 + 1 = 18 → write 8, carry 1
```

```
hundreds: 4 + 1 = 5
→ 585 ✓
```

Check 45 × 13: 45 × 10 + 45 × 3 = 450 + 135 = **585**. ✓

7. **The name.** *"Tirthaji (1965) calls this proportional scaling **Anurūpyeṇa** — 'proportionally.' The ×11 case is the special case k = 1, where the scaling is trivial."*

Activity (15 min) — Mixed drill

See `activities/activity-02-anurupyena.md`.

- 16 problems across ×12, ×13, ×14, ×17, ×19.
- Paired self-check.

Wrap-up (2 min)

- *"How does ×11 fit into the ×12–×19 family?"* (It's the k=1 special case.)
- Preview Day 7: *"What about ×23? ×47? When does the trick stop helping?"*

Student-facing content

> **The ×12 to ×19 family — *Anurūpyeṇa***
>
> For any $N = 10 + k$ (where k is 1 to 9), multiply a 2-digit number $ab \times N$ by: 1. Units = $b \times k$ (carry if ≥ 10). 2. Tens = $a \times k + b$ (plus any carry; carry if ≥ 10). 3. Hundreds = a (plus any carry).
>
> When $k = 1$ (i.e. ×11), this is the original "split and add" trick.
>
> Tirthaji (1965) names this scaling pattern ***Anurūpyeṇa*** — "proportionally."

Homework

- 12 problems mixing ×12, ×13, ×14: see worksheet attached. Self-check with long multiplication on any 4.

Differentiation

- **Tier up:** Try 3-digit × 13 problems. Use the right-to-left running-scan from Day 4 with the k-scaling from today.
- **Tier down:** Focus only on ×12 today. Build fluency before adding higher k.

Teacher notes

- *Anurūpyeṇa* is the pattern-naming step. Some students will object: *"This is just long multiplication done right-to-left."* They are correct. Honour the observation; reframe: *"The win is doing it MENTALLY in one pass, with a clear rhythm."*
- The procedural complexity grows quickly with k. ×17 and ×19 are harder than ×11 by a real margin. Calibrate the problem set accordingly.
- This day is genuinely harder than Days 1–5. If the class is shaky, slow down — skip the harder k values and revisit on Day 9.

Citation(s) used in this lesson

- Tirthaji (1965), sub-sūtra *Anurūpyeṇa*.
- *Microsoft Word - sutras1.pdf* — for the sub-sūtra list.

Day 7 — When the Trick Doesn't Help

Module: ×11 (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will identify which multiplications are well-suited to the *Ekādhikena*/*Anurūpyeṇa* family and which are not, and will articulate the conditions that make a different sūtra (*Nikhilam* or *Ūrdhva-tiryagbhyāṁ*) more appropriate.

Materials

- Whiteboard.
- 8-problem comparison worksheet (×11 trick vs. long multiplication vs. *Nikhilam*-style).

Lesson flow

Warm-up (5 min)

- Quick chorus: 10 problems from Day 6's ×12–×19 family.

Core (20 min) — Where the trick breaks down

1. **Try ×23.** *"Let's try the same proportional pattern with multiplier 23 = 20 + 3."*
    - For 14 × 23, would the trick scale?
    - (14)(20 + 3) = 280 + 42 = 322. Long way works.
    - The "proportional middle" pattern doesn't extend cleanly past 19 because the multiplier has two non-trivial digits.
2. **Why?** Recall the identity: ×(10 + *k*) → 100*a* + 10(*ak* + *b*) + *bk*. Clean because the "tens" part of the multiplier is **just 1**. For ×23, the algebra is:

```
(10a + b)(20 + 3)
    = 200a + 20b + 30a + 3b
    = 200a + 30a + 20b + 3b
    = (100·2 + 100·0 + ...) — no longer a tidy "outer-stays, inner-sums" pattern.
```

3. **The honest summary.**

    The ×11 trick (*Ekādhikena Pūrveṇa*) and its scaled cousins (*Anurūpyeṇa* for ×12 to ×19) work because the multiplier is *very close to 10*. Specifically, the multiplier must be 10 + *k* where *k* is a single digit.

For multipliers farther from 10 (×23, ×47, ×98), a DIFFERENT sūtra is the right tool — Tirthaji's *Nikhilam* ("all from 9, last from 10") for multipliers near 100, or *Ūrdhva-tiryagbhyāṁ* ("vertically and crosswise") for general multiplication.

4. **Connect to Module 9 / 11.** Future modules will cover *Nikhilam* and *Ūrdhva-tiryagbhyāṁ* in depth. Today we just note their existence.

5. **The bigger lesson.** *"Every mental-math shortcut has a domain. Knowing WHEN a trick applies is part of being a fluent mathematician. The standard long-multiplication algorithm always works; the shortcuts are faster only inside their domain."*

Activity (15 min) — Comparison drill

- 8 problems. For each, decide *before* solving: would you use (A) the ×11 family trick, (B) long multiplication, or (C) a different mental shortcut you can articulate?

```
1. 67 × 11          5. 84 × 23
2. 47 × 13          6. 99 × 99
3. 56 × 19          7. 25 × 12
4. 38 × 47          8. 102 × 11
```

- Pair work. Solve and discuss your method choice.

- Answers and "best method" given on the back.

Wrap-up (5 min)

- Class compiles "the ×11 family domain": *all 2-digit numbers × (11, 12, 13, …, 19)* — about 90 × 9 = 810 distinct problems. That's a significant mental-math territory.
- Preview Day 8: *"Tomorrow — where do these 16 sūtras come from? The honest history."*

Student-facing content

The domain of the ×11 trick

The "split and add" trick works for multipliers of the form **10 + *k*** where *k* is a single digit (so ×11 through ×19).

For other multipliers, use: - Long multiplication (always works). - *Nikhilam* "all from 9, last from 10" — when both numbers are near 100 (e.g. 98 × 97). - *Ūrdhva-tiryagbhyāṁ* "vertically and crosswise" — general multiplication.

Knowing **when to use** a trick is as important as knowing **how**.

Homework

- Find any 3 multiplication problems in your homework or a textbook. For each, decide which method (×11 family / long / other) you'd use and **why**. Write 1 sentence of justification per problem.

Differentiation

- **Tier up:** Find any 2-digit × 2-digit multiplication where the ×11 trick is NOT optimal but a different mental-math shortcut would beat long multiplication. (Examples: 25 × 24 = 25 × 24 = 600 via 25×4×6; 75 × 12 = 900 via 75×12 = 75 × 12.)
- **Tier down:** Focus on identifying ×11 vs. non-×11 cases. Skip the "best alternative method" question.

Teacher notes

- This day is short by design — it's the honesty hour. Don't over-fill.
- Some students will resist the framing *"the trick has a limited domain"* and try to force-fit ×23 into a pattern. Honour the impulse, but redirect: *"You'd be inventing your own technique — which is great, but it's not what this trick does."*
- Mention that Tirthaji's 16 sūtras collectively cover a wide territory; ×11 is just one of them. This sets up Day 8.

Citation(s) used in this lesson

- Tirthaji (1965), sūtra list (especially *Nikhilam* — sūtra 2, and *Ūrdhva-tiryagbhyāṁ* — sūtra 3).
- *Microsoft Word - sutras1.pdf* — full sūtra list.

Day 8 — Tirthaji's Sūtra System in Context

Module: ×11 (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will name at least 5 of Tirthaji's 16 sūtras, explain in 2–3 sentences when and by whom *Vedic Mathematics* was published, and state honestly that the "Vedic" attribution is academically contested.

Materials

- Printed handout: the list of 16 sūtras + 13 sub-sūtras (from `Microsoft Word - sutras1.pdf`).
- Optional: a paragraph excerpt from Tirthaji's preface (teacher's prepared synopsis).

Lesson flow

Warm-up (3 min)

- Recall: name the sūtra behind the ×11 trick. (*Ekādhikena Pūrveṇa*.)
- Name the sub-sūtra behind ×12–×19. (*Anurūpyeṇa*.)

Core (25 min) — The 16 sūtras and the provenance question

1. **The 16 sūtras.** Hand out the list. Walk through abbreviated:

#	Sanskrit	English gloss
1	*Ekādhikena Pūrveṇa*	"By one more than the previous"
2	*Nikhilam Navataścaramaṁ Daśataḥ*	"All from 9 and the last from 10"
3	*Ūrdhva-tiryagbhyāṁ*	"Vertically and crosswise"
4	*Parāvartya Yojayet*	"Transpose and apply"
5	*Śūnyaṁ Sāmyasamuccaye*	"When the sum is the same, that sum is zero"
6	*(Ānurūpye) Śūnyam Anyat*	"If one is in ratio, the other is zero"
7	*Sankalana Vyavakalanābhyām*	"By addition and by subtraction"
8	*Pūraṇāpūraṇābhyām*	"By completing or non-completing"
9	*Calana-Kalanābhyām*	"By differentiation and integration"
10	*Yāvadūnam*	"Whatever the extent of its deficiency"
11	*Vyaṣṭi-Samaṣṭiḥ*	"Specific and general"
12	*Śeṣānyaṅkena Caramena*	"The remainders by the last digit"
13	*Sopāntya-dvayam Antyam*	"The penultimate and the last"
14	*Ekanyūnena Pūrveṇa*	"By one less than the previous"
15	*Guṇitasamuccayaḥ*	"The product of the sum is the sum of the products"
16	*Guṇaka Samuccayaḥ*	"All the multipliers"

Plus 13 sub-sūtras (including *Anurūpyeṇa*).

2. **Who was Tirthaji?**
    - **Bharati Krishna Tirthaji** (1884–1960). Śaṅkarācārya of Govardhana Maṭha (Puri) — i.e. head of one of the four major Hindu monastic seats founded by Ādi Śaṅkara.
    - Studied at Madras Christian College. Multi-lingual scholar.
    - Wrote the manuscript of *Vedic Mathematics* sometime between roughly 1911 and 1918 according to its editor's foreword.
    - The manuscript was lost or unfinished for decades.
    - Tirthaji **died in 1960**.
    - The book was **published in 1965**, five years after his death, edited by Vasudev Sharan Agrawala.
3. **The provenance question.** *"Tirthaji claimed the 16 sūtras come from a Vedic text called the* Atharva-veda Pariśiṣṭa *— a supplement to the Atharva Veda."*
    - **The problem:** No other Sanskritist has produced a manuscript of this *Pariśiṣṭa* containing these 16 sūtras.
    - **The mathematician S. G. Dani** (TIFR, IIT Bombay) wrote a careful critique in 1993, reprinted in *Resonance* (October 2001), pointing out:
        - No manuscript evidence.

        - The Sanskrit style of the sūtras is more recent than Vedic.
        - The mathematical techniques have 19th–20th century parallels.
4. **Honest framing.** Two things can be true at once:

    1. The ×11 trick (and the rest of Tirthaji's 16-sūtra system) **works**. It is a useful, beautiful piece of mental arithmetic.
    2. The trick is **not verifiably ancient Vedic mathematics**. The best current evidence is that Tirthaji synthesised it in the early 20th century, drawing on classical Indian arithmetic + his own pedagogical genius, and named the sūtras in Sanskrit.

    The synthesis itself is a real Indian intellectual achievement. The "Vedic" naming is a 20th-century interpretive choice that scholars dispute.
5. **What's at stake.** If the sūtras are Vedic (3000+ years old), they're a major historical claim. If they're 20th-century synthesis, **they're still a real achievement** — just a different one. Tirthaji's pedagogical genius is not diminished by being recent.

Discussion (15 min) — Take a position

Pose two questions to the class. Take a brief show-of-hands; then take 4–5 student statements (no debate, just listening).

1. *"Does it matter whether the trick is 'ancient' or '20th-century'? Why or why not?"*
2. *"If it turns out to be 20th-century, should we still teach it as 'Vedic mathematics'? What should we call it?"*

End by noting: *"At Senior band you'll read Dani's critique in full and form a deeper position."*

Wrap-up (2 min)

- Preview Day 9: *"Tomorrow — mixed-practice day plus project work. Friday is the final assessment."*

Student-facing content

> **Tirthaji and the 16 sūtras**
>
> Bharati Krishna Tirthaji (1884–1960), Śaṅkarācārya of Govardhana Maṭha, wrote *Vedic Mathematics* between roughly 1911 and 1918. The book was published in 1965, five years after his death.
>
> Tirthaji claimed the 16 sūtras (including our *Ekādhikena Pūrveṇa*) come from a Vedic appendix-text. **No other Sanskritist has produced this manuscript.** S. G. Dani (1993) argues the sūtras are best understood as a 20th-century synthesis with Indian-mathematics roots, not a verifiably ancient Vedic teaching.
>
> The trick works. The history is contested. Both can be true. The Indian intellectual achievement is real either way.

Homework

- Read the 16-sūtra list. Pick ONE sūtra (other than the two we've studied) and find what mathematical operation it's used for. (Sūtras 2, 3, 4, 10, 14 are well-documented in Tirthaji 1965.) Write 3 sentences.

Differentiation

- **Tier up:** Read a one-paragraph excerpt from Dani (1993). What is Dani's strongest critique? What does he NOT criticise?
- **Tier down:** Memorise the names of sūtras 1, 2, 3, and 14 (the four most commonly taught). Skip the historicity discussion details.

Teacher notes

- Some students (and their families) believe firmly that the trick is "ancient Vedic mathematics." Honour the belief; don't dismiss it. Frame your position as "the academic consensus is more sceptical, and you'll read the evidence at Senior band."
- Avoid civilisational chest-thumping ("ancient Indians knew this before everyone"). The Indian achievement here is the synthesis and clarity of Tirthaji's pedagogical work; that's enough to celebrate without overreach.
- Avoid the opposite error too — dismissing the trick as "just Western math with Sanskrit labels." The pedagogical insight (the named patterns, the rhythm) is genuinely Tirthaji's.

Citation(s) used in this lesson

- Tirthaji, B. K. (1965). *Vedic Mathematics*. Motilal Banarsidass. — Sūtra list and preface.
- Dani, S. G. (1993). "Myths and Reality: On Vedic Mathematics." Reprinted in *Resonance* (October 2001).
- *Microsoft Word - sutras1.pdf* (SutraGanita corpus) — the 16+13 sūtra list.

Day 9 — Mixed Practice + Project Session

Module: ×11 (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will complete a mixed-practice drill covering 2-digit ×11, 3-digit ×11, and ×12–×19 problems, and will make substantial progress on their project (video, workbook, or proof poster).

Materials

- Mixed-practice drill sheet (20 problems).
- Project materials per pair: paper, markers, phone (for video pairs), template sheets (for workbook pairs).
- `assessments/rubric.md` posted on the wall for reference.

Lesson flow

Warm-up (3 min)

- 8 quick mental ×11 problems including carry. Should be near-instant.

Mixed-practice drill (12 min)

- Distribute 20-problem sheet:
  - 5 problems: 2-digit × 11 (no-carry)
  - 5 problems: 2-digit × 11 (carry)
  - 5 problems: 3-digit × 11
  - 5 problems: 2-digit × 12 through ×19
- Solo, untimed, but aim to finish in 10 min.
- 2 min: pair-swap, peer-check three problems each.

Project session (28 min)

- Project format options (pick on Day 8 or earlier):
  - **Format 1: 60-second teaching video** for a Grade 4 audience.
  - **Format 2: Printed workbook** — 25 problems organised in difficulty levels, with worked examples and an answer key.
  - **Format 3: Proof poster** — one A3 sheet showing the algebraic derivation of $(10a + b) \times 11 = 100a + 10(a+b) + b$ with examples and a 3-line explanation.
- Each pair shows the teacher their plan in the first 5 min. Teacher signs off or redirects.
- Work time. Teacher circulates:
  - Check video pairs have a quiet recording space.
  - Check workbook pairs have grid paper.
  - Check poster pairs have A3 sheets and at least 2 markers.
- Final 5 min: each pair shares their working title and ONE thing they're stuck on. Class offers one suggestion.

Wrap-up (2 min)

- *"Tomorrow is the final assessment and project share. Bring your finished artefact. Polish, don't restart."*

Student-facing content

Today's two jobs

1. **Drill.** 20 mixed problems. Self-assess.
2. **Project work.** Make real progress today — by end of class you should be 70% done.

Project Day 10: 2-minute presentation + 1-min Q&A. Be ready.

Homework

- Finish your project at home. 30 minutes max. If it's not finishable in 30 minutes, **simplify it** — better a complete simple project than a half-done ambitious one.

Differentiation

- **Tier up:** Add a "challenge" section to your project: a 4-digit × 11 worked example with cascading carries.
- **Tier down:** Use the workbook template provided. Don't custom-design from scratch today.

Teacher notes

- The project here is intentionally smaller-scope than Module 5's, because the technique is narrower (just ×11 plus the ×12–×19 family) and the assessment is more on the algebra than on the artefact.
- Watch for pairs who haven't started by mid-class. Pull them aside; offer the simplest format (proof poster) as a fallback.
- The proof poster format is recommended for pairs that struggled on Day 3 — making the algebra public forces them to internalise it.

Citation(s) used in this lesson

- (No new citations — application day.)

Day 10 — Final Assessment + Project Showcase

Module: ×11 (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will demonstrate mastery of the ×11 family of techniques via a summative quiz and a 2-minute project presentation.

Materials

- Summative quiz (`quizzes/summative.md`).
- Stopwatch.
- Rubric (`assessments/rubric.md`) per pair.
- Reflection slip per student.

Lesson flow

Warm-up (2 min)

- *"Quiz first (15 min), then presentations."*

Part 1: Summative Quiz (15 min)

- Distribute `quizzes/summative.md`. Closed notebook.
- Covers: 2-digit × 11 (with and without carry), 3-digit × 11, *Anurūpyeṇa* extension, the algebraic identity, the historicity question (1 short-answer).
- Collect at 15-min mark.

Part 2: Project Presentations (25 min)

- 2-min presentation + 1-min Q&A per pair (3 min total per pair).
- ~8 pairs fit in 25 min. Reserve next class for remaining pairs if needed.
- During each presentation:
  - Teacher marks rubric in real-time.
  - One assigned audience member asks one question.
  - Other audience members fill "I learned / I wonder" slips.

Wrap-up (3 min)

- Closing question (oral): *"What's one mental-math habit you'll keep using after this module?"*
- Reflection slip submitted (reproduced in `student-workbook.md`).

Student-facing content

> **Today: quiz + presentations**
>
> 15 min for the summative quiz (closed book).
>
> 2-minute presentation per pair, then 1-min audience question.
>
> **Reflection (due at end of class):** 1. What was the most surprising thing you learned about ×11? 2. Did your view of "Vedic mathematics" change during this module? How? 3. If you had three more days, what would you do differently?

Homework

- None. Module is complete.

Differentiation

- **Tier up:** Students finishing the quiz early can attempt a bonus "challenge" sheet — 4-digit × 11 plus a 2-digit × 19 mix.
- **Tier down:** Open-notebook quiz for students with documented learning needs.

Teacher notes

- Mark the quiz immediately after class while the presentations are fresh.
- Don't rush the reflection — the "did your view change?" question is the cultural-honesty payoff of the module.
- If presentations spill into Day 11, protect that time.

Citation(s) used in this lesson

- All sources verified across the module.

Writing Prompts

Writing Prompts — Module 7 Eleven Multiplication (Middle)

Six prompts. Useful as exit tickets, homework reflections, or project warm-ups.

Reflection prompts

1. (Day 1 or Day 5, end of class, 3 min) What did it feel like the first time you watched the teacher do 23 × 11 in 2 seconds? Describe what your mind did — disbelief, suspicion, "show me again"?

2. (Day 5 or Day 7, 5 min) You've now seen one Vedic-math trick in detail. Has your *feeling* about mental math changed since Day 1? Be honest — even if the answer is "no."

Explanatory prompts

3. (Day 3 or Day 4, homework, ~120 words) Explain the ×11 trick to a Grade 4 student who has never seen it. Use one specific example. No more than 6 sentences. Include the carry case in 1 sentence.

4. (Day 6 or homework, ~100 words) Explain the *Anurūpyeṇa* extension in your own words. Why does the same idea scale from ×11 to ×12, ×13, …, ×19? Use one example.

Argumentative prompt

5. (Day 8 or homework, 1 paragraph) Some people claim the ×11 trick is "ancient Vedic mathematics." Others say it's a 20th-century pedagogical technique published in 1965. Pick a position and defend it in 6–8 sentences. Cite at least one source.

Creative prompt

6. (Day 4 or Day 9, optional, ~150 words) Write a short script (50–150 words) in which a frustrated student and a confident ×11-trick user have a conversation. The frustrated student raises one good objection ("this is just multiplication, why a special name?"). The confident user gives a clear, honest answer. Let the conversation be real — no "and then they magically agreed" endings.

Prompt-to-band notes

- For **Primary** band, prompts 1 and 6 work as oral dialogues; prompt 3 works as a drawing.
- For **Senior** band, prompt 5 deserves a full essay with citations from Tirthaji (1965), Dani (1993), and Plofker (2009).

How to give feedback

- Prompts 1, 2: acknowledge with one sentence — no correction.
- Prompts 3, 4: respond with “the part that landed” and “the part that confused me.”
- Prompt 5: use the project rubric — does the writing acknowledge the other side, cite a source, reach a clear position?
- Prompt 6: no grading. Read the best ones aloud (with permission).

Homework

Homework — Module 7 Eleven Multiplication (Middle)

Day 1 → Day 2

Task: Find any five 2-digit numbers (with digit-sum < 10) around your house — phone numbers, page numbers, prices. Compute each × 11 using the trick. Bring the worked list.

Time: 10 min.

Day 2 → Day 3

Task: Eight mental ×11 problems including the carry case: 76, 84, 85, 88, 93, 96, 67, 78 — each × 11. Write only the answers. Time yourself; target under 30 seconds total.

Time: 5 min.

Day 3 → Day 4

Task: Apply the proof to a number you choose. Pick any 2-digit number, write it as $10a + b$, distribute by 11, and show the final form. Show all 5 lines of working.

Time: 10 min.

Day 4 → Day 5

Task: Solve 10 three-digit × 11 problems (on the back of today’s worksheet). Time yourself — goal: under 90 seconds total. Self-check three with long multiplication.

Time: 15 min.

Day 5 → Day 6

Task: 25-problem mixed drill sheet (handed out in class). Solo, untimed. Target: 100% correct.

Time: 15 min.

Day 6 → Day 7

Task: 12 problems mixing ×12, ×13, ×14 (worksheet attached). Self-check 4 with long multiplication.

Time: 15 min.

Day 7 → Day 8

Task: Find any 3 multiplication problems in your homework or a textbook. For each, decide which method (×11 family / long / other) you'd use and **why**. Write 1 sentence of justification per problem.

Time: 10 min.

Day 8 → Day 9

Task: Read the 16-sūtra list. Pick ONE sūtra (other than the two we've studied) and find what mathematical operation it's used for. Write 3 sentences. Confirm your project plan in writing (4–6 sentences).

Time: 20 min.

Day 9 → Day 10

Task: Finish your project. Polish, don't restart. Time your presentation in your head — must be 2 min.

Time: capped at 30 min.

Day 10 → end

Task: None. Module is complete. Optional: read ahead to Module 9 / 11 (Nikhilam, Ūrdhva-tiryagbhyāṁ) if curious.